# Chapter 10: Sequence Analysis

## 1. Introduction

Sequence analysis examines the temporal order of learning events to identify common patterns and trajectories. Distance metrics (Levenshtein, Longest Common Subsequence) quantify dissimilarity between sequences, enabling clustering of similar learner pathways and discovery of typical learning patterns.

## 2. Dataset & Variables

Event sequences from 25 students. Each sequence represents the temporal order of course interactions (e.g., forums, quizzes, assignments, videos, wikis). Mean sequence length: 14.3 events.

## 3. Methods

Two distance metrics were computed: Levenshtein distance (minimum edit distance) and Longest Common Subsequence (LCS) distance. Hierarchical clustering was applied using Ward linkage on the Levenshtein distance matrix. Sequences were grouped into 3 clusters based on dendrogram cut.

## 4. Results

**Sequence Lengths:**

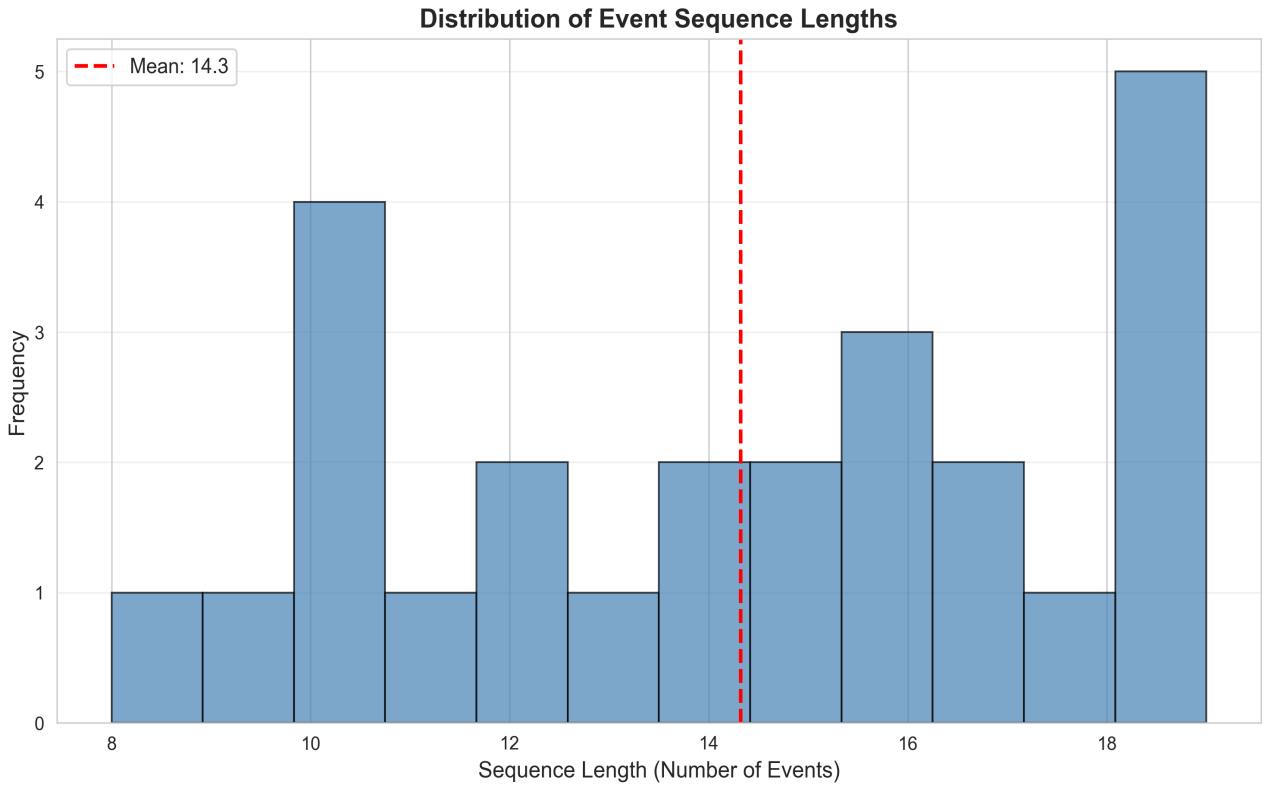


Fig 1: Distribution of event sequence lengths.

**Dendrogram:**

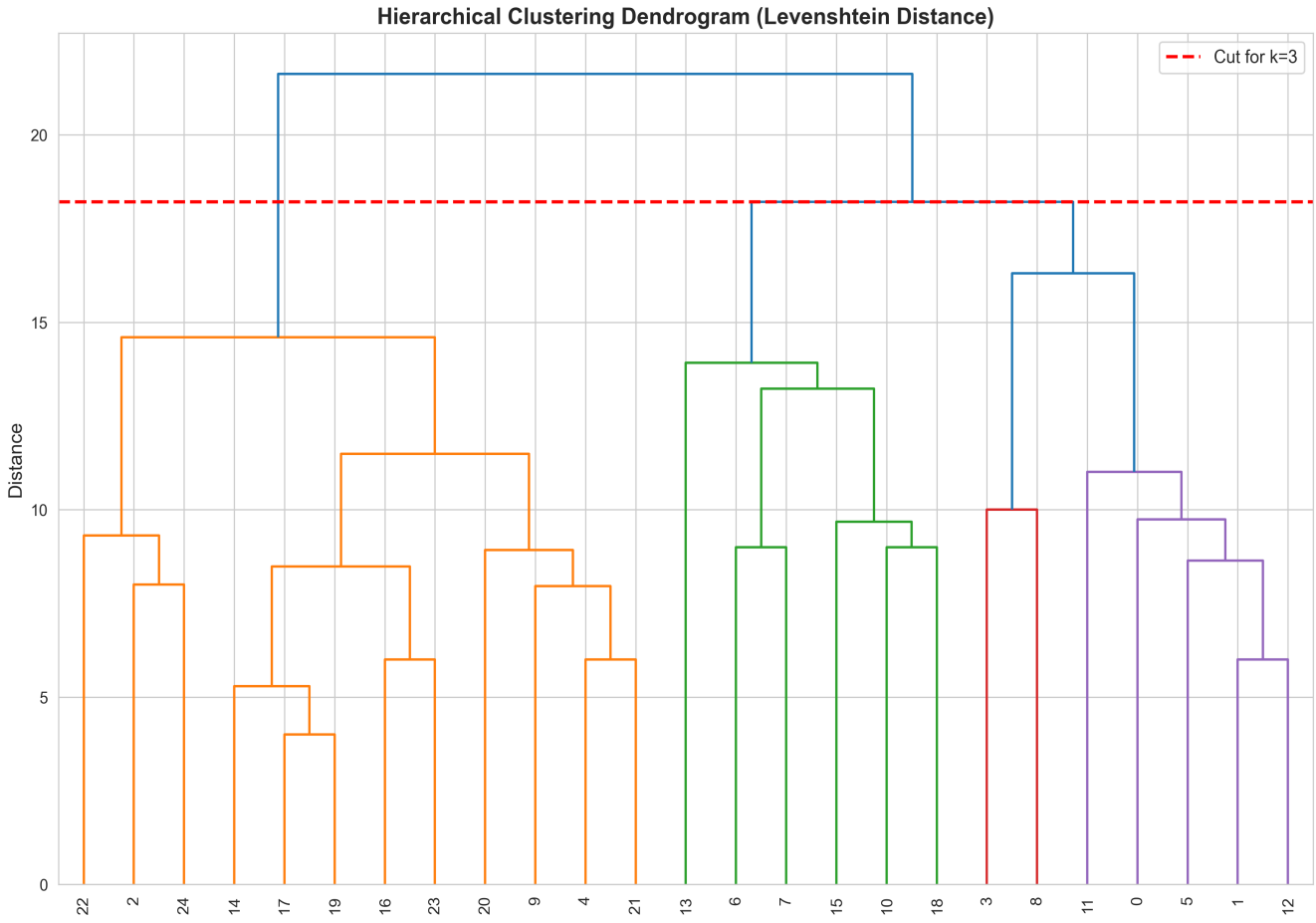


Fig 2: Hierarchical clustering dendrogram showing sequence relationships.

**Cluster Distribution:**

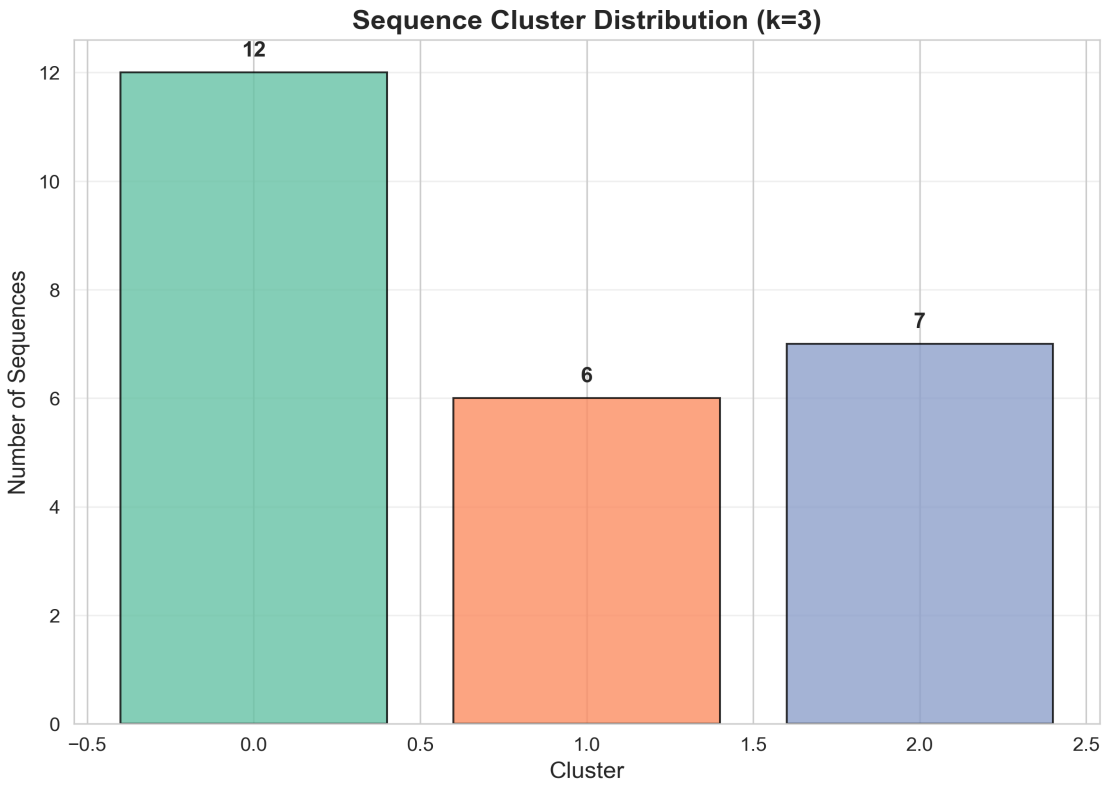


Fig 3: Distribution of sequences across clusters.

## 5. Interpretation

Sequence clusters reveal distinct learning pathway patterns among students. Similar sequences (low Levenshtein distance) indicate comparable engagement patterns. These patterns can inform instructional design decisions.

## 6. Limitations

Sequence analysis ignores timing information; absolute timestamps are not considered. Distance metrics may not capture domain-specific similarity between event types. Optimal cluster count determination is heuristic.

## 7. Reproducibility

Run cells sequentially from setup through PDF generation. Results are saved under `figures/` and `sequence_clustering_results.csv`.

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